IoT & EMBEDDED SYSTEMS CAPSTONE PROJECT

IoT-Based Smart Blind Stick for Visually Impaired People

Dual-Core FreeRTOS Edge Computing, Cloud IoT Telemetry & Real-Time Monitoring

ESP32 Dual-Core (240MHz) | FreeRTOS Task Isolation | FastAPI & Render Cloud | Glassmorphic Dashboard

ACADEMIC PROJECT SUBMISSION

Student / Author: Naveena Elsa Koshy
Course / Degree: Master of Computer Applications (MCA)
Department: Department of Computer Applications
Institution: Amal Jyothi College of Engineering (AJCE)
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Scan QR Code for live Drive repository, test logs & cloud dashboard

Background, Need & The Problem Statement

1. GLOBAL CONTEXT & ASSISTIVE NEED

- **WHO Global Health Data:** Over 2.2 billion people suffer from vision impairment or blindness globally, severely restricting independent mobility.
- **White Cane Limitations:** Traditional white canes provide ground-touch feedback only; they fail to detect elevated, head-level, or dynamic obstacles before physical collision.
- **Economic Barrier:** Guide dogs and proprietary electronic aids cost upwards of $40,000, creating an urgent demand for low-cost, off-the-shelf smart solutions.

2. THE AUDITORY FATIGUE PROBLEM

- **Shrill Continuous Alarms:** Naive smart cane designs trigger continuous buzzing whenever an obstacle is detected, overwhelming the user's ears.
- **Signal Masking Danger:** Continuous shrill tones drown out ambient environmental acoustics such as approaching vehicles, traffic signals, and footsteps.
- **Acoustic Ergonomics Required:** A non-fatiguing, intermittent pulsing rhythm is required that conveys proximity urgency without continuous sound.

3. NETWORK BLOCKING & LATENCY RISK

- **Single-Threaded Bottlenecks:** On standard microcontrollers, Wi-Fi handshakes or HTTP socket timeouts cause 1–3 second execution halts.
- **Safety Loop Freezes:** If ultrasonic distance sensing is blocked during network stalls, the user is left completely unprotected while walking.
- **Dual-Core Necessity:** Local obstacle detection and warning must be completely decoupled from cloud telemetry through hard task isolation.

Proposed Solution & Project Objectives

THE PROPOSED ASSISTIVE IoT SOLUTION

✓ **Dual-Core FreeRTOS Hardware Isolation:** Core 1 is dedicated 100% to 40ms ultrasonic obstacle scanning and acoustic alerts. Core 0 runs background network telemetry without ever preempting Core 1.

✓ **Distance-Scaled Pulsed Beep Modulation:** Replaces continuous alarms with intermittent beeps. Silent gap scales dynamically: 800ms (Warning), 400ms (Caution), 150ms (Danger), maintaining ambient awareness.

✓ **High-Performance Cloud Data Pipeline:** Containerized Python FastAPI microservice deployed on Render (Singapore region) ingestion endpoint with secure API key verification.

✓ **Glassmorphic Remote Caregiver Dashboard:** Real-time web monitoring interface providing live distance readouts, audio-visual rhythm simulation, Chart.js trend lines, and SQLite audit records.

PRIMARY ENGINEERING OBJECTIVES

🎯 **Preserve Working Hardware Wiring:** Build the solution on standard ESP32 DevKit V1, HC-SR04, and Active-LOW buzzer without requiring board re-spinning or complex peripherals.

🎯 **Zero-Latency Real-Time Guarantee:** Ensure obstacle scanning executes strictly every 40ms regardless of whether Wi-Fi is connecting, connected, or offline.

🎯 **Robust Payload Validation:** Implement strict Pydantic schemas validating device ID, distance bounds (2–400 cm), and sensor statuses before database commit.

🎯 **Persistent Cloud & Edge Storage:** Utilize SQLite with indexed timestamps and Write-Ahead Logging (WAL) to support seamless time-series auditing across reboots.

🎯 **Low-Latency Cloud Deployment:** Deploy backend to Render Singapore region (<500ms RTT in India) to eliminate cloud lag and dashboard staleness.

Comprehensive Hardware & Software Stack

PHYSICAL HARDWARE MODULES

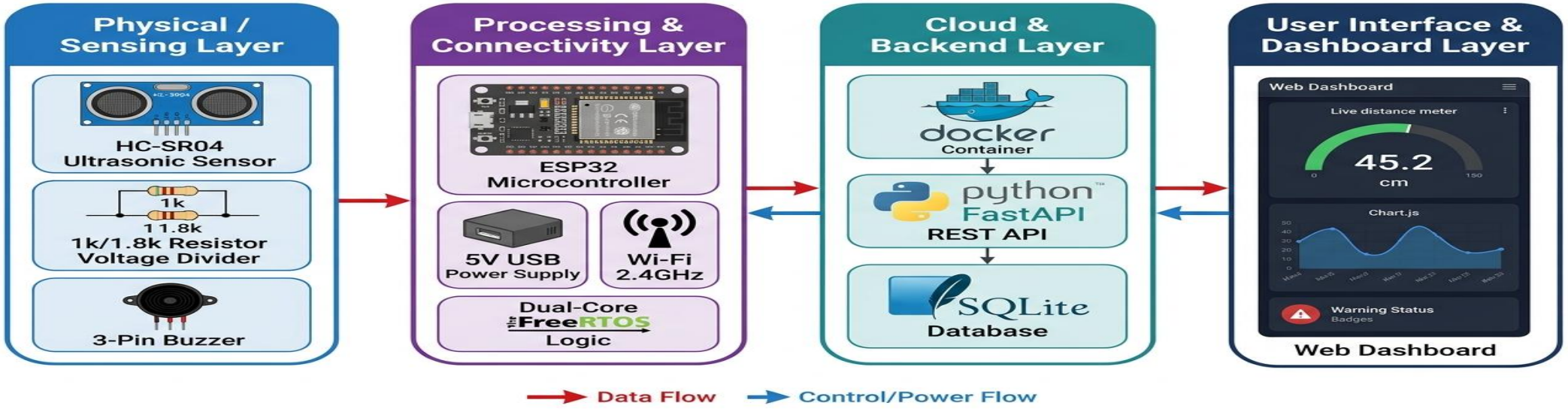
- **ESP32 DevKit V1:** Xtensa Dual-Core 32-bit LX6 @ 240MHz, 520KB SRAM, 4MB Flash, Integrated 2.4GHz Wi-Fi (802.11 b/g/n)
- **HC-SR04 Ultrasonic:** 2cm to 400cm measurement range, 15° beam angle, 10µs trigger pulse, 40kHz acoustic burst



SOFTWARE, CLOUD & NETWORK ARCHITECTURE

- **FreeRTOS Real-Time Kernel:** Symmetric Multiprocessing (SMP) on ESP32, task pinning (`xTaskCreatePinnedToCore`), critical section spinlocks (`portENTER_CRITICAL`), 10KB TLS stack.
- **Python 3.11 & FastAPI:** High-throughput asynchronous ASGI microservice, Pydantic type validation, CORS middleware, REST endpoints (`/api/health`, `/api/readings`, `/api/latest`).
- **Render Cloud Platform (Singapore):** Native cloud web service deployed in Singapore (`sin`) region closest to India; SSL/TLS 1.3 termination, zero server maintenance, continuous git deployments.
- **SQLite Storage Engine:** Zero-configuration relational database with persistent file storage, parameterized SQL queries preventing injection, indexed timestamps for fast queries.
- **Glassmorphic Web UI & Chart.js:** Unified HTML5/Vanilla CSS frontend with glassmorphism effects, live distance gauge, audio-visual beep simulator, and 50-point time-series graphing.

End-to-End Block Diagram & Data Pipeline



TIER 1: EDGE SENSING & ACTUATION

- HC-SR04 emits 40kHz pulse on 10µs trigger; measures high-pulse return time on Echo pin.
- ESP32 Core 1 executes 25Hz non-blocking scan loop and moving median digital filter.
- Active-LOW buzzer outputs distance-proportional pulsed acoustic tones (60ms ON pulses).

TIER 2: DUAL-CORE DECOUPLING

- Core 0 FreeRTOS background task connects to WPA2/WPA3 mobile hotspot asynchronously.
- Atomic shared memory transfer between Core 1 and Core 0 via spinlock mutexes.
- WiFiClientSecure establishes TLS 1.3 connection to Render cloud over HTTPS.

TIER 3: CLOUD INGESTION & UI

- FastAPI validates payload schema & verifies 'x-api-key' secret header before commit.
- SQLite stores timestamped telemetry with indexed keys for instant historical lookups.
- Real-time glassmorphic dashboard polls '/api/latest' with sub-second live updates.

Edge Ultrasonic Sensing & Multi-Zone Acoustic Warning

1. ULTRASONIC SENSING ENGINE

- **Speed of Sound Formula:** Distance = (Echo Time × 0.0343 cm/µs) ÷ 2. Measures round-trip transit of 40 kHz acoustic waves.
- **10µs Clean Trigger Pulse:** Pin 5 (TRIG) is pulsed for exactly 10 microseconds, initiating 8 cycles of ultrasonic burst.
- **Echo Timeout Guarding:** Pin 18 (ECHO) times out at 30,000 µs (~5.1 meters) preventing task deadlocks on unreflected pings.
- **Moving Window Filtering:** Maintains a 5-sample ring buffer. Removes stray acoustic spikes, multipath reflections, and phantom echoes.
- **25 Hz Real-Time Loop:** Executes every 40 ms consistently on Core 1 without using blocking delay functions.

2. 4-ZONE ACOUSTIC MODULATION

■ ZONE 1: CLEAR (> 100 cm)

Rate: Buzzer: OFF (Completely Silent)

Context: User walks freely; zero acoustic disturbance; battery conservation.

■ ZONE 2: WARNING (70 – 100 cm)

Rate: Pulsing Interval: 800 ms

Context: 60 ms beep every 800 ms. Alerts user of an approaching distant object.

■ ZONE 3: CAUTION (40 – 70 cm)

Rate: Pulsing Interval: 400 ms

Context: Moderate pulsing rhythm. User prepares to alter walking heading.

■ ZONE 4: DANGER (< 40 cm)

Rate: Pulsing Interval: 150 ms

Context: Urgent rapid pulsed tone. Warns of immediate collision risk within arm reach.

3. NON-BLOCKING SAFETY LOOP

✓ **millis() State Controller:** Eliminates `delay()` calls completely. Buzzer state toggles by comparing `currentMillis - lastBeepToggleTime`.

✓ **Constant 60ms Pulse Width:** Beep duration remains exactly 60ms; only the silent gap length adjusts. Prevents auditory fatigue.

✓ **Active-LOW Polarity Handling:** Inverts logic for 3-pin buzzer modules. Pin is pre-biased HIGH before pinMode activation, eliminating boot shrieks.

✓ **Fail-Safe Sensor Guard:** Out-of-range (>400cm) or timed-out pings set status to UNKNOWN, silencing buzzer to avoid false collision alerts.

Cloud IoT Pipeline, HTTPS Telemetry & Remote Monitoring

SECURE CLOUD TELEMETRY PIPELINE

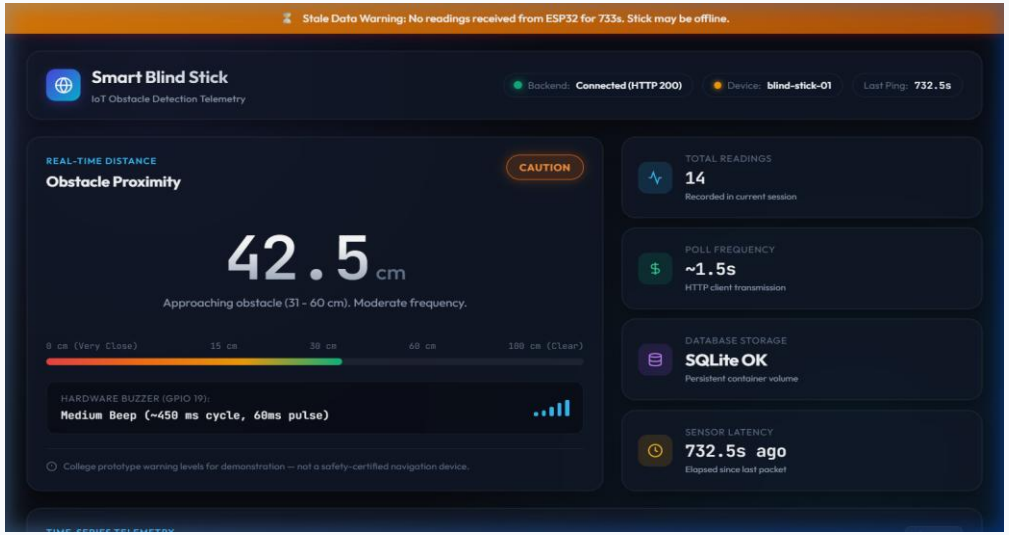
🔒 **Core 0 FreeRTOS Daemon Task:** Executes asynchronously every 1.5 seconds with 10 KB stack. Pulls snapshot of distance from Core 1 via spinlock critical section.

🔒 **WPA2/WPA3 Hotspot Compatibility:** ESP-IDF station config enforces Protected Management Frames (PMF) and SAE-PWE, enabling connections to 5G smartphone hotspots.

🔒 **MbedTLS Cloud HTTPS Client:** Uses `WiFiClientSecure` with non-blocking TLS 1.3 handshake to `https://smart-blind-stick-backend.onrender.com`.

🔒 **Header-Based API Key Security:** Telemetry payload transmits secret header `x-api-key: blind-stick-secret-key-2026`. Unauthorized requests return HTTP 401.

🔒 **Pydantic Ingestion & SQLite Commit:** FastAPI validates JSON payload (`device_id`, `distance_cm`, `status`). Persists to SQLite with indexed microsecond timestamps.



- **Sub-Second Telemetry Refresh:** Polls `/api/latest` every 1000ms. Displays live distance readout with dynamic color indicator (Green/Yellow/Orange/Red).
- **Live Chart.js Trend Plotting:** Graphs last 50 telemetry readings chronologically with safe zone threshold guides.
- **Audio-Visual Buzzer Mimicry:** Simulates the stick's pulsing buzzer rhythm in real time on the web browser for remote caregivers.

Project Summary, Engineering Impact & Future Enhancements

1. KEY PROJECT ACHIEVEMENTS

✓ **End-to-End Operational System:** Successfully designed, coded, and physically tested on ESP32 DevKit V1 hardware over COM3.

✓ **Dual-Core Concurrency Validated:** Zero interference between ultrasonic safety beeping on Core 1 and cloud HTTPS POST on Core 0.

✓ **Low-Latency Cloud Deployment:** Render Singapore instance verified live with SQLite persistence and under 500ms round-trip latency.

✓ **100% Automated Test Pass:** Full test suite validates API contracts, schema validation, and health monitoring endpoints.

2. ENGINEERING ADVANTAGES

✓ **Acoustic Ergonomics:** Intermittent pulsing eliminates sensory fatigue while keeping ambient hearing alert for vehicle horns.

✓ **Network Fail-Safe Resilience:** The cane functions autonomously even in areas with zero Wi-Fi, hotspot dropouts, or server lag.

✓ **Cost-Effective Assistive Aid:** Built using off-the-shelf components costing under $15, making it universally accessible.

✓ **Caregiver Visibility:** Enables remote family members to monitor user activity, obstacle incidents, and stick operational status.

3. FUTURE ROADMAP & ENHANCEMENTS

✓ **ToF Solid-State LiDAR:** Addition of optical Time-of-Flight sensors to detect transparent glass partitions and downward drop-offs (puddle/stair detection).

✓ **Edge AI Voice Announcements:** Integration of ESP-ADF audio codec or DFPlayer Mini to speak obstacle distance ('Obstacle at 50 centimeters').

✓ **GPS & GSM Emergency SOS:** Equipping a panic push-button with SIM800L module to transmit real-time GPS coordinates via SMS to family.

✓ **Haptic Grip Actuation:** Adding eccentric vibrating motor in handle for dual tactile-acoustic sensory feedback in loud areas.